

Далее проанализируем
от 05.09
~7.

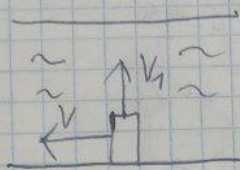
Дано:

$$V = 3 \frac{\text{км}}{\text{ч}}$$

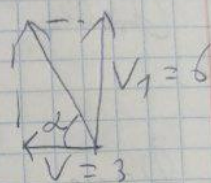
$$V_1 = 6 \frac{\text{км}}{\text{ч}}$$

2-?

Решение:



=>



Решение: 60°

~2.

Дано:

$$S_1 = 4 \text{ км} \quad t_0 = 40 \text{ мин} = 40 \cdot \frac{60}{60} \text{ ч}$$

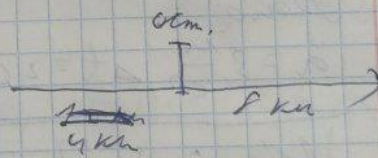
$$V_1 = 12 \frac{\text{км}}{\text{ч}}$$

$$V_2 = 8 \frac{\text{км}}{\text{ч}}$$

$$S_2 = 8 \text{ км}$$

$V_{\text{ср}} - ?$

Решение:



$$t_1 = \frac{4}{12} = \frac{1}{3} \text{ ч}$$

$$\text{в среднем } t_0 = 2400 \text{ с}$$

$$\text{в среднем } t_0 = \frac{40}{60} = \frac{2}{3} \text{ ч}$$

среднее время

$$t_2 = \frac{s_2}{v_2} = \frac{p}{f} = 12$$

$$v_g = \frac{4+8+0}{1+\frac{2}{3}+\frac{1}{3}} = \frac{12}{2} = 6 \frac{km}{h}$$

Answer: $v_g = 6 \frac{km}{h}$
~ 3.

Dans:

$$v_0 = 9 \frac{m}{s} \quad t = 300$$

$$a = 0,5 \frac{m}{s^2}$$

$$v_1 = ?$$

Formule:

$$v_1 = v_0 + at =$$

$$= 9 + 30 \cdot 0,5 = 19 \frac{m}{s}$$

Answer: $19 \frac{m}{s}$

~ 4.

Dans:

$$a = 3 \frac{m}{s^2}, \quad \Delta t = 20$$

$$\Delta v = ?$$

Formule:

$$v(t) = at = 3 \cdot 20 = 60 \frac{m}{s}$$

Answer: $60 \frac{m}{s}$

Danu:

$$V_0 = 20 \frac{m}{s}$$

$t = ?$

Jawab:

$$g = 9,8 \frac{m}{s^2}$$

$$V_{max} = 0, V_{max} = V_0 - g t \Rightarrow$$

$$\Rightarrow 20 - 9,8 t = 0$$

$$9,8 t = 20$$

$$t = \frac{20}{9,8} \approx 2,04 \text{ s}$$

Jawab: $\approx 2 \text{ s}$

~ 6.

Danu:

$$V_0 = 0; h = 50 \text{ m}$$

$$t = 3 \text{ s} \quad g = 9,8 \frac{m}{s^2}$$

~~?~~ $y = ?$

Jawab:

$$y = y_0 + V_0 t + \frac{g t^2}{2}$$

$$\text{diketahui, maka } y_0 = h:$$

$$y = 50 - \frac{9,8 \cdot 9}{2} \approx 5,9 \text{ m}$$

Jawab: $5,9 \text{ m} (\approx 6 \text{ m})$

~ 8.

Danu:

$$V_1 = 60 \frac{km}{h}$$

$$V_2 = 40 \frac{km}{h}$$

$V_{cp} = ?$

Jawab:

$$\text{diketahui, maka } V_1 = V_2 = 2 \text{ s}$$

$$t_1 = \frac{1}{60}; t_2 = \frac{1}{40}$$

$$V_{cp} = \frac{\frac{2}{60} + \frac{2}{40}}{\frac{2}{60} + \frac{2}{40}} = \frac{2 \text{ s}}{\frac{2}{60} + \frac{2}{40}} = \frac{240 \text{ s}}{5} = 48 \frac{km}{h}$$

~~avg $\frac{10,8 \cdot 10}{36} = 2,97 \cdot 10^2$~~

$$V_0 = \frac{81 \cdot 1000}{3600} = 22,5 \frac{m}{s}$$

$$V_1 = \frac{10,8 \cdot 10}{36} = 3 \frac{m}{s}$$

$$a = \frac{\Delta V}{\Delta t} = -\frac{2}{f} = -0,25 \frac{m}{s^2}$$

Graviti: $-9,81 \frac{m}{s^2}$

ad.

Dato:

$$V_1 = 16 \frac{km}{h}$$

$$V_2 = 12 \frac{km}{h}$$

$$V_3 = 5 \frac{km}{h}$$

$$V_{eq} = ?$$

Timeline:

$$s_1 = \frac{s}{2}; t_2 = \frac{t - t_1}{2}; t_3 = t_2$$

$$s_1 = s_2 + s_3$$

$$V_{eq} = \frac{s}{t} = \frac{s}{t_1 + t_2 + t_3} = \frac{s}{t_1 + 2t_2}$$

$$t_1 = \frac{s_1}{V_1} = \frac{s}{2V_1} \Rightarrow$$

$$s_2 = V_2 t_2; s_3 = V_3 t_2$$

$$\Rightarrow s_1 = V_2 t_2 + V_3 t_2 = t_2 (V_2 + V_3) \Rightarrow$$

$$\Rightarrow t_2 = \frac{s_1}{V_2 + V_3}$$

$$V_{eq} = \frac{s}{\frac{s}{2V_1} + \frac{2s_1}{V_2 + V_3}} = \frac{s}{\frac{s}{2V_1} + \frac{2s}{2(V_2 + V_3)}} =$$

Onbren: $4 \frac{\text{km}}{\text{h}}$

~9.

Dans:

$$V_1 = 6 \frac{\text{km}}{\text{h}} \quad t_1 = \frac{2}{3} \text{ h}$$

$$V_2 = 4 \frac{\text{km}}{\text{h}} \quad t_2 = \frac{1}{3} \text{ h}$$

$V_g = ?$

Remarque:

$$S_1 = 6 \cdot \frac{2}{3} = 4 \text{ km}$$

$$S_2 = \frac{4}{3} \text{ km}$$

$$V_g = \frac{\frac{4 + \frac{4}{3}}{\frac{2}{3} + \frac{1}{3}} = \frac{16}{3} = 5,3 \frac{\text{km}}{\text{h}}$$

$$= \frac{16}{3} = 5,3 \frac{\text{km}}{\text{h}}$$

Onbren: $5,3 \frac{\text{km}}{\text{h}}$

~10.

Dans:

$$V_0 = 18 \frac{\text{km}}{\text{h}} \quad V_1 = 10,8 \frac{\text{km}}{\text{h}} \quad \Delta t = 1 \text{ h}$$

$a = ?$

Remarque:

$$a = \frac{\Delta V}{\Delta t} = \frac{10,8 - 18}{1} = -7,2 \frac{\text{km}}{\text{h}^2}$$

$$a = \frac{\Delta V}{\Delta t} = \frac{10,8 - 18}{1} = -7,2 \frac{\text{km}}{\text{h}^2}$$

Onbren: $10,8 \frac{\text{km}}{\text{h}}$

$V_1 = 10,8 \frac{\text{km}}{\text{h}}$

$a = \frac{\Delta V}{\Delta t}$

Onbren:

Dans:

$$V_1 = 10,8 \frac{\text{km}}{\text{h}}$$

$$V_2 = 10,8 \frac{\text{km}}{\text{h}}$$

$$V_3 = 5,4 \frac{\text{km}}{\text{h}}$$

$V_g = ?$

$V_g =$

Дано:

$$\omega R = \text{const}$$

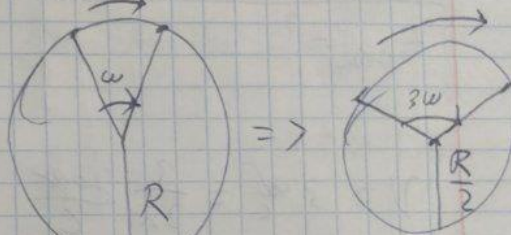
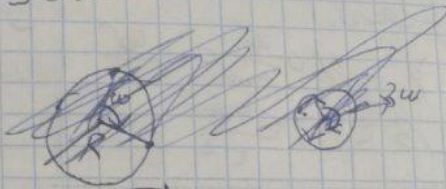
$$R_0 = R \quad R_1 = \frac{R_0}{2}$$

$$\omega R_0 = \omega \quad \omega_1 = 3\omega$$

$$\frac{a_{n2}}{a_{n1}} = ?$$

~3.

Решение:



$$a_{n1} = \frac{v_1^2}{R_1}; \quad a_{n2} = \frac{v_2^2}{R_2}$$

$$v_1 = \omega R; \quad v_2 = \frac{3\omega R}{2} = 1,5\omega R$$

$$\frac{a_{n2}}{a_{n1}} = \frac{\frac{(1,5\omega R)^2}{R/2}}{\frac{(\omega R)^2}{R}} =$$

$$= \frac{2 \cdot 2,25 \omega^2 R^2 \cdot R}{R \cdot \omega^2 R^2} = 4,5$$

Ответ: отношение 4,5 раз.